

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and maintained **Laboratory Information Systems (LIS)** with continuous improvement cycles, evolving platform functionality through **PHP** back-end web application development, **database creation and modification** for new analytical features, and **API design** to integrate instrument data streams, processing **40% increased data throughput** across multi-site cloud deployments.
- Engineered **data import/export systems** and **instrument-to-system data transfer pipelines** using **Python** scripting and **SQL** (PostgreSQL, MySQL, Oracle) to automate **data quarantine systems** and quality control verifications, reducing manual data validation time by **60%** while maintaining audit-trail compliance and **statistical analysis** of measurement precision and uncertainty.
- Implemented **cloud computing infrastructure** and **on-premise troubleshooting** frameworks across **AWS** (EC2, S3, Lambda, CloudWatch) and **Docker** containerization, supporting **document release deployment** and **change management** procedures that aligned with quality management system scope and regulatory requirements across distributed laboratory environments.
- Designed **Web Portal and Quality Management System** architecture, leading cross-functional meetings with analytical, field services, quality, finance and sales teams to document business process automation requirements, translate data requirements into **efficient, straightforward processes**, and deliver **design documents** describing system procedures with **reusable code libraries** for future feature development.
- Optimized applications for **ease of use, speed and scalability** through performance tuning and logical code layout following **script procedures**, enabling team members to independently extend functionality; produced comprehensive **documentation** and **user guides** for Web Portal deployment and **ongoing maintenance** across **multi-site-based applications**.
- Collaborated with experienced developers to review company requirements with Managers, propose **design solutions**, and keep pace with latest technology developments to make informed recommendations on product direction and feature prioritization; demonstrated strong **time management** and **problem-solving skills** while developing programming expertise and gradually assuming greater technical responsibility.

SKILLS

Web & Back-End Development – PHP, Java Spring Boot, REST API design, FastAPI, Flask.

Database & Data Systems – SQL (PostgreSQL, MySQL, Oracle), database design and modification, data modelling, data warehousing, ETL pipelines.

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, cloud computing infrastructure, multi-site deployments.

Data Management & Quality Control – Data import/export systems, data quarantine systems, automated quality control verifications, statistical analysis, measurement precision and uncertainty, bias controls.

DevOps & Release Management – CI/CD (Jenkins pipelines), Git, change management, document release deployment, Linux, shell scripting.

Development Practices – Agile/Scrum, code reviews, design documentation, reusable code libraries, logical code layout, script procedures.

Business & Domain – Laboratory Information Systems (LIS), Quality Management Systems, analytical process automation, cross-functional requirements gathering, technical communication.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India

01/2021

PROJECTS

Laboratory Data Pipeline & Quality Management Portal (HCL Technologies)

- Architected and deployed a **Laboratory Information System (LIS) data integration platform** using **PHP** back-end services and **REST APIs** to ingest high-volume instrument data streams, manage **data import/export workflows**, and enforce **automated quality control verifications**, eliminating **80% of manual data quarantine review cycles** while maintaining statistical audit trails for measurement precision and uncertainty analysis.
- Designed **multi-database schema** across **PostgreSQL, MySQL, and Oracle** to support **data quarantine systems** with real-time status flags, implement **automated validation rules** for analytical results, and enable **cross-site data synchronization** via **Python ETL scripts** deployed on **AWS Lambda**, processing **50,000+ instrument readings daily** across 12 laboratory locations.
- Built **Web Portal** front-end with intuitive dashboards and **Quality Management System** reporting capabilities using **reusable code libraries**, enabling field services, quality, and analytical teams to independently access lab data, track method validations, and generate compliance-ready reports; reduced data retrieval time from **2 hours to 5 minutes** through optimized query design.
- Implemented **cloud computing infrastructure** on **AWS** with **Docker containerization** and **on-premise fallback systems**, documented **deployment procedures** and **change management protocols**, and delivered runbooks for troubleshooting; achieved **99.7% uptime** across production and development environments serving 200+ laboratory staff.
- Collaborated with cross-functional teams (analytical, quality, finance) to gather business process requirements, authored comprehensive **design documents** and **user guides**, and led UAT coordination for phased rollout; mentored 2 junior developers on **script procedures**, code organization standards, and **API development best practices**.

Production Data Monitoring & Incident Response Automation (HCL Technologies)

- Developed **automated quality control verification systems** and **data management workflows** using **Python** scripting and **SQL** to monitor system health, detect anomalies in instrument data feeds, and trigger remediation workflows, reducing incident response time by **65%** and enabling **24/7 on-premise and cloud troubleshooting** without manual intervention.
- Built **scalable data pipelines** with **performance optimization** for ease of use and speed, leveraging **AWS CloudWatch** and **Docker** to handle **3x data throughput growth**; documented **logical code layout** and procedures enabling team members to extend and maintain systems independently with zero post-deployment issues.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)